

MONT JOLI, QC, Canada

WMO: 717180

Lat: 48.603N

Lon: 68.210W

Elev: 52

StdP: 100.70

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.0	(c) -20.6	(d) -28.3	(e) 0.3	(f) -22.7	(g) -25.9	(h) 0.4	(i) -20.2	(j) 16.0	(k) -8.0	(l) 14.0	(m) -9.5	(n) 6.1	(o) 270	(p) 0.704

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.9	(c) 27.0	(d) 20.0	(e) 25.2	(f) 18.9	(g) 23.7	(h) 18.0	(i) 20.9	(j) 25.4	(k) 19.8	(l) 24.0	(m) 18.8	(n) 22.7	(o) 5.5	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.1	23.7	18.1	13.1	22.6	17.1	12.3	21.4	60.9	25.3	57.2	24.1	53.8	22.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.9	(b) 11.4	(c) 10.2	(e) -25.9	(f) 30.9	(g) 3.0	(h) 1.7	(i) -28.1	(j) 32.1	(k) -29.8	(l) 33.1	(m) -31.5	(n) 34.0	(o) -33.7	(p) 35.2		
(4) 12.9	(5) 11.4	(6) 10.2	(7) -26.2	(8) 23.1	(9) 2.9	(10) 1.3	(11) -28.3	(12) 24.0	(13) -30.1	(14) 24.8	(15) -31.7	(16) 25.5	(17) -33.8	(18) 26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-10.8	-9.7	-4.6	2.2	9.0	14.7	18.1	17.3	12.9	6.5	0.5	-6.2
	DBStd	10.93	6.21	5.66	5.25	3.91	3.88	3.90	3.06	3.04	3.85	3.90	4.55	5.16
	HDD10.0	2879	644	551	453	234	66	4	0	0	13	123	288	502
	HDD18.3	5223	903	784	711	483	289	122	42	57	169	366	536	760
	CDD10.0	774	0	0	0	1	36	146	251	226	99	15	2	0
	CDD18.3	77	0	0	0	0	1	13	34	24	5	0	0	0
	CDH23.3	412	0	0	0	0	13	82	172	117	27	1	0	0
	CDH26.7	61	0	0	0	0	2	16	24	17	3	0	0	0
Wind	WSAvg	4.8	5.7	5.5	5.4	5.0	4.5	4.1	3.9	3.9	4.2	4.8	5.4	5.6
Precipitation	PrecAvg	969	88	71	72	59	82	72	78	90	85	81	83	102
	PrecMax	1301	211	206	156	137	153	167	174	205	154	134	190	152
	PrecMin	758	36	11	24	12	18	15	38	30	26	31	41	42
	PrecStd	118	38	36	37	30	34	36	28	41	35	30	32	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	6.8	10.4	16.5	25.2	28.5	29.1	28.7	26.2	20.9	17.2	7.7
		MCWB	5.8	5.4	7.4	10.4	16.1	19.6	21.2	20.6	19.9	15.6	14.2	5.8
	2%	DB	3.3	2.3	6.1	12.4	21.4	25.6	27.0	26.2	23.4	17.7	11.9	4.4
		MCWB	2.1	0.8	3.3	7.3	13.8	18.2	20.2	19.7	18.2	13.6	9.4	3.1
	5%	DB	0.7	0.4	4.1	9.7	18.1	23.4	25.2	24.4	21.2	14.8	8.9	2.5
		MCWB	-0.3	-0.4	1.9	5.5	12.4	17.2	19.1	18.7	16.9	11.5	6.4	1.2
	10%	DB	-1.9	-1.2	2.3	7.5	15.6	21.5	23.7	22.7	19.2	12.5	6.4	0.8
		MCWB	-2.8	-2.1	0.4	4.4	10.6	16.0	18.2	17.7	15.5	9.7	4.5	-0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	5.4	8.4	11.5	17.8	21.4	22.8	21.9	21.1	16.5	14.3	6.2
		MCDB	7.1	6.7	10.1	15.4	23.6	26.9	27.1	27.2	25.0	19.9	16.6	7.3
	2%	WB	2.2	1.2	3.7	7.9	15.2	19.5	21.2	20.4	18.9	13.9	9.8	3.3
		MCDB	3.3	1.9	5.4	11.2	19.5	23.9	25.4	25.0	22.3	17.3	11.7	4.2
	5%	WB	-0.3	-0.3	2.0	6.1	13.0	18.0	20.0	19.5	17.4	11.9	6.7	1.4
		MCDB	0.7	0.3	3.8	9.1	17.7	21.9	24.1	23.5	20.8	14.6	8.3	2.4
	10%	WB	-2.7	-2.0	0.6	4.6	11.1	16.7	18.9	18.5	15.7	10.0	4.7	0.0
		MCDB	-1.8	-1.3	2.0	7.0	14.9	20.6	22.9	22.0	18.9	12.1	6.1	0.7
Mean Daily Temperature Range	5% DB	MDBR	7.3	7.3	7.1	7.1	8.9	9.3	8.9	8.7	8.2	6.4	5.5	5.9
		MCDBR	10.2	7.9	8.8	10.8	13.0	11.9	10.6	10.7	10.8	10.1	9.1	7.8
		MCWBR	9.8	7.2	7.1	7.4	7.6	7.0	5.9	6.2	7.1	7.3	7.8	7.4
	5% WB	MCDBR	10.0	7.5	7.9	10.2	12.1	10.7	9.9	9.6	10.5	9.7	8.7	7.7
		MCWBR	9.7	7.0	6.8	7.6	7.7	7.0	6.5	6.2	7.6	7.5	7.9	7.4
Clear-Sky Solar Irradiance	taub	0.261	0.269	0.300	0.359	0.367	0.388	0.414	0.393	0.367	0.329	0.298	0.271	
	taud	2.341	2.312	2.286	2.287	2.352	2.352	2.314	2.374	2.426	2.510	2.460	2.364	
	Ebn at Noon	815	894	916	889	894	873	843	846	835	817	766	751	
	Edh at Noon	68	88	108	121	118	119	122	110	95	74	62	59	
All-Sky Solar Radiation	RadAvg	1.25	2.17	3.34	4.40	5.14	5.49	5.37	4.61	3.52	2.07	1.11	0.83	
	RadStd	0.13	0.24	0.24	0.38	0.45	0.40	0.54	0.30	0.24	0.17	0.12	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	+1.16	+1.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page